

Margherita Calderan, Ph.D.

My research interests include computational models of cognition, Bayesian statistics, and metascience.

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🐙 github.com/mar-cald 🏠 gitlab.pavlovia.org/mar.cald 📄 ResearchGate 🏠 OSF

Current Position

Research Assistant

May 2026 - Present

Department of Developmental Psychology and Socialization, University of Padova, Italy

- Number-space association: A comparative, developmental and neurobiological approach
- PI: Prof. Silvia Benavides

Research

Research Fellow

Oct. 2024 - Mar. 2026

Department of General Psychology, University of Padova, Italy

- Tails or types? Testing the dimensional hypothesis in neurodevelopmental disorders
- Supervisor: Prof. Enrico Toffalini

Education

Ph.D. in Psychological Sciences - cum laude

Defense: Mar. 2025

University of Padova, Italy

- Thesis: Balancing stability and flexibility in cognitive control and learning: the effects of arousal
- researchdata.cab.unipd.it/1437
- Supervisor: Prof. Roberta Sellaro, Co-Supervisor: Prof. Nicola Cellini

Postgraduate Internship

Jan. 2021 - Dec. 2021

Psychophysiology Lab, University of Padova, Italy

- Investigated the role of sleep characteristics in false memory formation and emotional reactivity.
- Supervisors: Prof. Andrea Spoto, Prof. Nicola Cellini

M.Sc. in Cognitive Neuroscience and Clinical Neuropsychology - 110/110 cum laude

Oct. 2018 - Dec. 2020

University of Padova, Italy

- Thesis: Sleep characteristics in adults with high functioning autism: a polysomnographic study
- Supervisor: Prof. Nicola Cellini

Curricular Internship

Dec. 2019 - Aug. 2020

Psychophysiology Lab, University of Padova, Italy

- Analyzed polysomnographic data to examine sleep architecture in young adults with high-functioning autism.
- Collaborated with the University of Trento and the University of Tübingen.
- Supervisor: Prof. Nicola Cellini

B.Sc. in Cognitive Psychology and Psychobiology - 109/110

Sept. 2015 - Jul. 2018

University of Padova, Italy

- Thesis: Gender differences in the long-term consolidation of emotional memories
- Supervisor: Prof. Michela Sarlo

Abroad

Visiting Scholar

1.5 months (Sept. 2025 - Oct. 2025)

Department of Biomedical Data Sciences, Leiden University Medical Center, the Netherlands

- Research on Bayesian approaches to the problem of multiplicity.
- Supervisor: Prof. Jelle Goeman

Visiting Ph.D. Student

4 months (Feb. 2024 - Jun. 2024)

Cognitive Psychology Unit, University of Leiden, the Netherlands

- Worked on a research project investigating how mind-wandering affects cognitive flexibility, applying computational models (psyarxiv.com/quhns) to analyze behavioral data.
- Supervisor: Dr. Bryant Jongkees

Research Intern

2 months (Jun. 2019 - Aug. 2019)

Donders Sleep and Memory Lab, Donders Institute, the Netherlands

- Assisted with a project on the association between affective memory bias, medial prefrontal schema processing, and sleep in depressed patients and healthy controls.
- Assisted with a project on lucid dream induction.
- Supervisor: Prof. Martin Dresler

Erasmus +

5 months (Jan. 2017 - Jun. 2017)

Jaume I University, Spain

- Completed a semester-long academic exchange as part of my undergraduate studies.

Publications

Published

1. Feraco, T., **Calderan, M.** & Pellegrino, G. (2025). The Living SEB Skills Project: A Living Resource for Review and Meta-Analysis of Social, Emotional, and Behavioral Skills. *European Journal of Personality*. doi:10.1177/08902070261467477.
2. Mazzoni, N., Kurz, E.V., Cellini, N., Ciringione, L., **Calderan, M.**, Gallitto, G., Falco, S., Venuti, P., Zinke, K., Born, J., & Caria, A. (2026). Aperiodic and periodic neural activity during sleep in autism spectrum disorders. *BMC Medicine*. doi:10.1186/s12916-026-04771-6.

3. **Calderan, M.**, Visalli, A., Degasperi, G., Toffoli, L., Sellaro, R., & Cellini, N. (2026). Predictions under sleep restriction: The effects of partial sleep deprivation on learning and inhibitory control. *Royal Society Open Science*. doi:10.1098/rsos.252345.
4. **Calderan, M.**, Jongkees, J.B., Gambarota, F., Pastore, M., Cellini, N., & Sellaro, R. (2025). Isolating working memory gating processes under stress. *Collabra::Psychology*. doi:10.1525/collabra.140750.
5. **Calderan, M.**, Toffalini, E., Mirandola, C., & Cellini, N. (2025). Sleep does not affect false memory production on a script-based test. *Neuropsychologia*. doi:10.1016/j.neuropsychologia.2025.109210.
6. Toffoli, L., **Calderan, M.**, Del Popolo Cristaldi, F., Duma, G. M., Calcagni, A., Pastore, M., Tarantino, V., & Mento, G. (2025). Can I afford one more candy? How motivational contexts shape adaptive cognitive control in children. *Developmental Psychology*. doi:10.1037/dev0001976.
7. Sobol, M., Błachnio, A., **Calderan, M.**, Degasperi, G., Mioni, G., Hryhorchuk, I., Plucińska, E., Stasiniewicz, J., Kwapiński, J., Jankowski, K., & Cellini, N. (2025). Balanced time perspective as a factor mitigating the negative consequences of sleep disruptions in the perinatal period. *Chronobiology International*, 1-12. doi:10.1080/07420528.2025.2490499.
8. **Calderan, M.**, & Visalli, A. (2024). Challenges of meta-learning and rational analysis in large worlds. *Behavioral and Brain Sciences*, 47, e148. doi:10.1017/S0140525X24000128.

Under Review

1. **Calderan, M.**, Gambarota, F., Toffalini, E., & Altoè, G., Adjusting the Probability of Direction for Multiple Testing. **Preprint:** doi:10.31234/osf.io/8zwx2_v2.
2. Barbiero, J., Zeilinga, K., **Calderan, M.** & Mioni, G., The contribution of the cerebellum in time perception: A systematic review and meta-analyses.
3. Stefanelli, G., Toffoli, L., Del Popolo Cristaldi, F., Duma, G.M., **Calderan, M.**, ... & Mento, G., Growing Up in a Changing World: volatility-driven implicit associative learning throughout development.
4. Visalli, A., Calistrone, F. M., **Calderan, M.**, Donnarumma, F., Zorzi, M., & Ambrosini, E., A computational framework for integrating Predictive processes with evidence Accumulation Models (PAM). **Preprint:** doi:10.48550/arXiv.2411.13203.
5. Toffalini, E., **Calderan, M.**, Pagan, R., Tobia, V., & Sonuga-Barke, E., Tails or Types? A Critical Systematic Review of Taxometric Studies of Neurodevelopmental Conditions. **Preprint:** osf.io/ys5ad/files/bem36.

Preregistration

1. Manente, M., Altoè, G., Alfarone, I., **Calderan, M.** (2026, July 14). Mapping Researcher Degrees of Freedom in Meta-Analyses of Randomized Controlled Trials of EMDR for PTSD: A Systematic Scoping Review. osf.io/8hefb.
2. Toffalini, E., **Calderan, M.**, Sità, L., Feraco, T., & Gambarota, F., Link Functions and Generalized Linear Models in Interaction Testing within Psychological Research. osf.io/sjdu4.
3. Sirota, M., Šrol, J., Lisi, M., Juanchich, M., Guglani, K., Silverstein, P., ... **Calderan, M.**, ... Schmidt, K. (2025, Jan. 14). Mapping and Increasing Error Correction Behaviour in a Culturally Diverse Sample. **Preprint:** doi:10.17605/OSF.IO/WN4VX.

In Preparation

1. Toffoli, L., **Calderan, M.**, Stefanelli, G., Duma, G.M., Del Popolo Cristaldi, F., Duma, G., Pastore, M., Frisina, S., Tarantino, V., & Mento, G., I am talking to you! Engagement modulates the effects of digital content exposure on preschoolers' adaptive cognitive control.

2. Bastianello, T., **Calderan, M.**, Loschiavo, A.G.M., Nascimben, C., Visibelli, E., Reoyo-Serrano, N., Altoè, G., & Benavides-Varela, S., Domestic Media Exposure Delays Attainment of Babbling in Infants While Adult Input Accelerates Progress.
3. Van Dooren, R., **Calderan, M.**, Sellaro, R., & Jongkees, J.B., Dynamic and contextual modulation of constraints on cognitive control.

Published Abstract

1. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. (2024). Restless Sleep, Uncertain Minds: Learning and Inhibitory Control Under Partial Sleep Deprivation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. escholarship.org/uc/item/65b2f0n7.
2. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. (2024). The Effects of Partial Sleep Deprivation on Adaptive Cognitive Control. *Sleep Medicine*. doi:10.1016/j.sleep.2023.11.156.
3. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. (2024). Expecting the unexpected: a closer look at the effect of partial sleep deprivation on learning and inhibitory control. *Journal of Sleep Research*. doi:10.1111/jsr.14291.
4. Baldassarri, A., Colombo, C., Bergamo, D., **Calderan, M.**, Varanini, A., Ferrentino, A., D'Atri, A., & Cellini, N. (2024). Exploring the impact of oscillatory events during a nap on emotional memory consolidation. *Journal of Sleep Research*. doi:10.1111/jsr.14291.

Conferences

1. Barbolan, S., Slaby, R., **Calderan, M.**, Knedlíková Wanková, V., & Bertamini, M. Emotional Responses to Renaissance Art: Differences Between Venetian and Bohemian Paintings. **Poster** accepted at the AIP - Experimental Section XXXII Conference, Venice (Italy), Sept. 2026.
2. **Toward Robust Replicability: Modern Perspectives and Methods**. Organizer: **Calderan, M.** Speakers: Alongi, E., Garofalo, S., **Calderan, M.**, Andreella, A. Moderator: Epifania, M.O. **Symposium** accepted for the AIP - Experimental Section XXXII Conference, Venice (Italy), Sept. 2026.
3. Barbolan, S., Slaby, R., **Calderan, M.**, Knedlíková Wanková, V., & Bertamini, M. Emotional Responses to Renaissance Painting: A Comparison Between Venetian and Bohemian Works. **Poster** accepted at the Visual Science of Art Conference (VSAC) 2026, Vision Forward: Perception, Art & Technology, London (United Kingdom), Aug. 2026.
4. Bastianello, T., Loschiavo, A., Nascimben, C., Reoyo-Serrano, N., **Calderan, M.**, Altoè, G., & Benavides-Varela, S. The effects of TV and electronic devices on language: a longitudinal study. **Poster** accepted at the ICIS 2026, Panama, Jul. 2026.
5. Bastianello, T., **Calderan, M.**, Loschiavo, A., Nascimben, C., Visibelli, E., Reoyo-Serrano, N., Altoè, G., & Benavides-Varela, S. L'esposizione ai media rallenta il raggiungimento del babbling e l'input dell'adulto ne accelera il progresso. **Talk** at CLASTA, Torino (Italy), Jun. 2026.
6. Barbolan, S., Slaby, R., **Calderan, M.**, Knedlíková Wanková, V., & Bertamini, M. Emotional and Aesthetic Experience in Venetian and Bohemian Paintings from the Renaissance. **Poster** presented at the International Association of Empirical Aesthetics (IAEA) 2026 Conference, Future Follows Past – Creating a Gesamtkunstwerk together for Arts and Aesthetics Research, Jena (Germany), May 2026.
7. Stefanelli G., Toffoli L., **Calderan M.**, Del Popolo Cristaldi F., Duma G. M., Tarantino V., Visalli A., Sellaro R., Mento G. Apprendimento associativo implicito e volatilità ambientale nell'ADHD. **Video-Poster** at Giornate di aggiornamento sull'uso degli strumenti in Psicologia Clinica dello Sviluppo, Bologna (Italy), Feb. 2025.

8. Stefanelli G., Toffoli L., **Calderan M.**, Duma G. M., Del Popolo Cristaldi F., Tarantino V., Visalli A., Sellaro R. & Mento G. Super Learning Bros: Developmental trajectories of probabilistic learning in volatile environment. **Talk** at the Cognitive Science Arena, Brixen (Italy), Feb. 2025.
9. **Calderan, M.**, Visalli, A., Degasperi, G., Calistrone, F.M., Sellaro, R., & Cellini, N. Expecting the unexpected: a closer look at the effect of partial sleep deprivation on learning and inhibitory control. **Poster** presented at Sleep Europe 2024, Seville (Spain), Sept. 2024.
10. **Calderan, M.**, Visalli, A., Degasperi, G., Calistrone, F.M., Sellaro, R., & Cellini, N. Restless Sleep, Uncertain Minds: Learning and Inhibitory Control Under Partial Sleep Deprivation. **Poster** presented at CogSci 2024: Dynamics of Cognition, Rotterdam (Netherlands), Jul. 2024.
11. Degasperi, G., **Calderan, M.**, Mioni, G., Meisner, M., Blachnio, A., Hryhorchuk, I., Plucinska, E., Sobol, M., & Cellini, N. The role of sleep, circadian rhythms, balanced time perspective on perinatal depression symptoms. **Poster** at the Cognitive Science Arena, Brixen (Italy), Feb. 2024.
12. Toffoli, L., **Calderan, M.**, Duma, G. M., Frisina, S., Tarantino V., & Mento, G. I'm talking to you! Interactive digital content effects on children's adaptive cognitive control. **Talk** at the Cognitive Science Arena, Brixen (Italy), Feb. 2024.
13. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. The Effects of Partial Sleep Deprivation on Adaptive Cognitive Control. **Poster** presented at World Sleep Congress, Rio de Janeiro (Brazil), Oct. 2023.
14. **Calderan, M.**, Visalli, A., Sellaro, R., & Cellini, N. The sleep-deprived predictive brain: modeling the effect of partial sleep deprivation on adaptive cognitive control. **Talk** at the XXIX Conference of the Italian Association of Psychology (AIP) - Experimental Psychology Section, Lucca (Italy), Sept. 2023.
15. **Calderan, M.**, Cellini, N., Sellaro, R. Stressed Minds: Unraveling Acute-Stress Effects on Working Memory Processes. **Mini-Talk** at the Cognitive Science Arena, Brixen (Italy), Feb. 2023.
16. Toffoli, L., **Calderan, M.**, Del Popolo Cristaldi, F., Duma, G. M., Calcagni, A., Pastore, M., Tarantino, V., & Mento, G. Can I afford one more candy? How motivational contexts shape adaptive cognitive control in children. **Talk** at the Cognitive Science Arena, Brixen (Italy), Feb. 2023.
17. **Calderan, M.**, Cellini, N., Sellaro, R. Exploring the Influence of Stress on Working Memory Gating Mechanisms. **Mini-Talk** at the XXX Conference of the Italian Association of Psychology (AIP), Padova (Italy), Sept. 2022.
18. **Calderan, M.**, Cellini, N., Sellaro, R. Understanding stress-induced Effects on Working Memory gating mechanisms. **Poster** presented at the 22nd meeting of the European Society for Cognitive Psychology (ESCoP), Lille (France), Aug. 2022.

Pre-conference workshops

- **Generalized Linear Models for Psychology: Understanding, Simulating, and Planning Your Analyses.** Pre-conference Workshop Cognitive Science Arena, Brixen (Italy). Feb. 2026 (4 hrs.). Hands-on methodological workshop held with Filippo Gambarota, PhD. psicostat.github.io/glm-workshop
- **A Practical Introduction to Evidence Accumulation Models in Visual Perception Research.** European Conference on Visual Perception, Mainz (Germany). Aug. 2025 (2.5 hrs.). Hands-on methodological workshop held with Carolina Maria Oletto, PhD. mar-cald.github.io/eam-workshop/eam-home.html.
- **Introduction to Reproducible Science.** Pre-School workshop *Replicability Crisis in Science*, Padova (Italy). Ed. 2025 and 2026 (2 hrs.). Material: github.com/Mar-Cald/repro-pre-school.

Invited talks

- **Bayesian inference and multiple comparisons: From idealization of methods to conscious practice.** Cognitive Neuroscience Seminars, Department of General Psychology, University of Padova, Italy. Dec. 2025.
- **From Cognitive Psychology to Bayesian Multiple Testing.** Medical Statistics Meeting, Department of Biomedical Data Sciences, Leiden University Medical Center, Leiden, The Netherlands. Oct. 2025.
- **Bayesian Multiple Testing.** STAT-Talk Series, Methodology and Statistics Department, Institute of Psychology, Leiden University, Leiden, The Netherlands. Sept. 2025.

Awards and Grants

- **Travel Grant:** Cognitive Science Arena, Brixen (Italy).
- **Travel Grant:** European Conference on Visual Perception, Mainz (Germany).
- **Travel Grant:** qHelp international seminar on Quantitative Psychology, Arrifana (Portugal).
- **Painter Award:** Best Poster, Cognitive Science Arena, Brixen (Italy).
- **Erasmus Placement Grant (2 Months):** Donders Sleep and Memory Lab, Nijmegen (The Netherlands).
- **Erasmus Placement Grant (5 Months):** Jaume I University, Castellón de la Plana (Spain).

Peer Reviews

Ad hoc reviewer (4 reviewed papers) for:

- Italian Journal of Applied Statistics (1)
- Frontiers in Psychology - Theoretical and Philosophical Psychology (1)
- Frontiers in Psychology - Cognition (1)
- PLoS ONE (1)

Teaching

[planned] means that the course has been officially scheduled but not delivered yet.

Lectures on contract (docenze a contratto)

- **Inferenza e statistica Bayesiana** (10 hours). Master in Quantitative Psychology, Department of Philosophy, Sociology, Education and Applied Psychology, University of Padova. Jul. 2026 [planned].
- **Modelli lineari** (10 hours). Master in Quantitative Psychology, Department of Philosophy, Sociology, Education and Applied Psychology, University of Padova. Jun. 2026 [planned]. Material: mar-cald.github.io/psqt/modelli-lineari.html
- **Test per le organizzazioni** (21 hours, 3 CFU). 2nd semester of the 2025/2026 academic year. Lecturer (titolare di modulo) for the Master's Degree in Psicologia Sociale, del Lavoro e della Comunicazione, University of Padova. Material: ottaviae.github.io/test-organizzazioni-fisppa
- **Introduction to meta-analysis with application in R** (5 hours, 1 CFU). Professor for the course within the PhD program in Psychological Sciences, University of Padova. Apr. 2026. Material: mar-cald.github.io/meta-phd

Seminars

- **Introduction to Bayesian Statistics with R and Stan** (30 hrs.). Intensive course on Bayesian statistics organized by the Italian Psychological Association (AIP). CEUB, Bertinoro (Italy), Jun. - Jul. 2026 [planned].
- **Introduction to R**. An intensive R programming course offered within the ARCA (Applied Research Course Academy) courses, Department of Developmental Psychology and Socialization, University of Padova. Material: github.com/arca-dpss/course-R-2425
 - Dec. 2024 (20 hrs.)
 - Apr. 2025 (20 hrs.)
 - Nov. 2025 (20 hrs.)
 - Mar. 2026 (20 hrs.)
- **Learning in Dynamic Environments: A Tentative Implementation of the Volatile Kalman Filter in Stan**. Presentation (1 hr.) on a Bayesian-based implementation of a learning algorithm for the 2023/24 Psicostat meeting series: psicostat.dpss.psy.unipd.it/meetings/slides/2024-03-22_calderan.pdf

Integrative teaching (Didattica integrativa)

- **Assistance** during lectures for the course **Testing Psicologico** (Prof. Tatiana Marci), Bachelor's degree in Scienze psicologiche dello sviluppo, della personalità e delle relazioni interpersonali, University of Padova (14 hrs.).
- Preparation and correction of **practical group exercises** and exams for the course **Experimental Research Methods in Psychology** (Prof. Roberta Sellaro), Master's degree in Developmental and Educational Psychology, University of Padova (14 hrs.).
- **Assistance** in the preparation of exercises and corrections for the course **General Psychology** (Prof. Enrico Toffalini), Bachelor's degree in Techniques and Methods in Psychological Sciences, University of Padova (15 hrs.).
- **Assistance** during lectures and preparation of **practical group exercise** for the course **General Psychology** (Prof. Giovanna Mioni), Bachelor's degree in Social and Work Psychological Sciences, University of Padova (14 hrs.).

Workshops

- **Bayesian Approaches To Multiplicity**. Psicostat HandZone on Bayesian Approaches To Multiplicity, University of Padova, Padova (Italy). Dec. 2025 (1.5 hrs.). psicostat.dpss.psy.unipd.it/pages/handzone.html
- **Tempi di risposta**. Workshop on reaction time distributions, University of Padova, Padova (Italy). Jun. 2025 (1 hr.). Material: mar-cald.github.io/tempi, psicostat.dpss.psy.unipd.it/pages/handzone.html

Academic Roles

- **Expert (Cultore della Materia) of Psychometrics (PSIC-01/C ex M-PSI/03)**. Department of Developmental Psychology and Socialization, University of Padova, Italy. Jan. 2026 - Present.
- **Representative of Research Fellows** in the Department of General Psychology, University of Padova. Feb. 2025 - Mar. 2026
- **Co-supervision of students** for their internships and Bachelor's (2) and Master's theses (1)

Organizational Activities

Methodological School: Replicability Crisis in Science

Member of the **Organizing team**. The school covers the epistemological, philosophical, and statistical foundations of replicability (psicostat.github.io/replicability-psy-unipd-phd).

- Jun. 2025, Padova (Italy)
- May 2026, Padova (Italy)
- **Lecturers:** Prof. Branden Fitelson, Prof. Giovanni Parmigiani, Dr. Filippo Gamarota, Dr. Margherita Calderan (ed. 2026)

International Winter School - The 4 Ms: Meta-research, Measurement, Multi-lab, and Multiverse data analysis in Behavioral Sciences

Member of the **Organizing team**. The Winter School aims to introduce attendees to advancements in meta-research, providing both theoretical insights and practical tools to conduct rigorous research. Feb. 2025, Padova (Italy) (winterschool4ms.dpg.unipd.it).

- **Lecturers:** Prof. Daniël Lakens, Prof. Dorothy Bishop, Prof. Jessica Flake, Prof. Lisa DeBruine, Prof. Ulrich Schimmack, Prof. Carlo Umiltà, Prof. Ettore Ambrosini, Dr. Filippo Gamarota, Dr. Giulia Calignano, Prof. Massimo Grassi, Prof. Livio Finos

ONIRICAMENTE

Member of the **Organizing team** of the one-day workshop dedicated to public outreach on the Psychology of Sleep. Jun. 2024, Padova (Italy) (sites.google.com/view/oniricamente/home-page?authuser=0).

Cognitive Neuroscience of Sleep Summer School

Member of the **Supporting team**. The Summer School examined methods for investigating sleep and memory in humans and animals. It also covered topics related to sleep medicine and dreaming. Nijmegen (Netherlands).

Third Mission (Terza Missione)

Replicability Crisis in Science - Methodological Seminars

2025

University of Padova, Italy

- The school aims to provide a broad and interdisciplinary perspective on replicability, along with tools that help participants explore aspects most relevant to their own field. Lectures cover the epistemological, philosophical, and statistical foundations of replicability and the so-called replication crisis (Jun. 2025).

Science4all

2025

University of Padova, Italy

- Centra il bersaglio: tra misurazione, probabilità e incertezza. Activity organized by the Psicostat Research Group and proposed during the Science4All activity at the University of Padova (Sept. 2025)

ONIRICAMENTE

University of Padova, Italy

2024

- One-day workshop dedicated to public outreach on the Psychology of Sleep (Jun. 2024)

Research Groups Membership

Psicostat

University of Padova

2024 - Present

- I am a core team member of Psicostat, an interdisciplinary research group specializing in quantitative psychology, psychometrics, psychological testing, and statistics.
- psicostat.dpss.psy.unipd.it/people.html

SleePy

University of Padova

2019 - Present

- I am a member of SleePy, a research group focusing on sleep psychology.

Collaborations

Prof. Jelle Goeman

Leiden University Medical Center, The Netherlands

2025 - Present

- Research on Bayesian approaches to the problem of multiplicity.

Prof. Aldo Solari

Ca' Foscari University of Venice, Italy

2025 - Present

- Research on Bayesian approaches to the problem of multiplicity.

Prof. Giovanni Parmigiani

Harvard University

2025 - Present

- Research and teaching projects about replicability in Psychology and Biomedical sciences.

Prof. Branden Fitelson

Northeastern University

2025 - Present

- Research and teaching projects about replicability in Psychology and Biomedical sciences.

Dr. Bryant Jongkees

Leiden University, The Netherlands

2024 - Present

- Research on stability and flexibility in cognitive control.

Dr. Antonino Visalli

University of Modena and Reggio Emilia, Italy

2023 - Present

- Research on computational models of learning and control.

Involvement in Research Groups

National Project on Malpractice in Psychological Research

Collaborative initiative aimed at identifying and addressing **methodological malpractice** in psychological science, with Enrico Toffalini (University of Padova), Filippo Gambarota (University of Padova) and Tommaso Feraco (University of Padova). My contributions include:

- Toffalini, E., **Calderan, M.**, Feraco, T., & Gambarota, F. Link Functions and Generalized Linear Models in Interaction Testing within Psychological Research. osf.io/sjdu4.
- Toffalini, E., **Calderan, M.**, Pagan, R., Tobia, V., & Sonuga-Barke, E. Tails or Types? A Critical Systematic Review of Taxometric Studies of Neurodevelopmental Conditions. Under review in *JCPP Advances*. github.com/EnricoToffalini/Taxometrics-Neurodevelopmental-Disorders/tree/main.

National Project on Adaptive Cognitive Control in Children

Part of a national research group focused on investigating adaptive cognitive control in typical, at-risk, and atypical populations within a Neuroconstructivist framework, with Giovanni Mento (University of Padova), Fiorella Del Popolo Cristaldi (University of Padova), Lisa Toffoli (University of Padova), Giulia Stefanelli (University of Padova), Gian Marco Duma (Epilepsy and Clinical Neurophysiology Unit, IRCCS E. Medea). The group has contributed to multiple peer-reviewed publications, conference presentations, and posters. My contributions specifically focused on the application of **Bayesian methods**, including **generalized multi-level models** and **survival analysis** techniques, in the following selected works:

- Toffoli, L., **Calderan, M.**, Del Popolo Cristaldi, F., Duma, G. M., Calcagni, A., Pastore, M., Tarantino, V., & Mento, G. (2025). Can I afford one more candy? How motivational contexts shape adaptive cognitive control in children. *Developmental Psychology*. Advance online publication. doi:10.1037/dev0001976.
- Stefanelli G., Toffoli L., **Calderan M.**, Del Popolo Cristaldi F., Duma G. M., Tarantino V., Visalli A., Sellaro R., Mento G. Apprendimento associativo implicito e volatilità ambientale nell'ADHD. **Video-Poster** at Giornate di aggiornamento sull'uso degli strumenti in Psicologia Clinica dello Sviluppo, Bologna (Italy), Feb. 2025.
- Stefanelli G., Toffoli L., **Calderan M.**, Duma G. M., Del Popolo Cristaldi F., Tarantino V., Visalli A., Sellaro R. & Mento G. Super Learning Bros: Developmental trajectories of probabilistic learning in volatile environment. **Talk** at the Cognitive Science Arena, Brixen (Italy), Feb. 2025.
- Toffoli, L., **Calderan, M.**, Duma, G. M., Frisina, S., Tarantino V., & Mento, G. I'm talking to you! Interactive digital content effects on children's adaptive cognitive control. **Talk** at the Cognitive Science Arena, Brixen (Italy), Feb. 2024.
- Toffoli, L., **Calderan, M.**, Del Popolo Cristaldi, F., Duma, G. M., Calcagni, A., Pastore, M., Tarantino, V., & Mento, G. Can I afford one more candy? How motivational contexts shape adaptive cognitive control in children. **Talk** at the Cognitive Science Arena, Brixen (Italy), Feb. 2023.

Training

Model-based Neuroscience Summer School

Jul. - Aug. 2024

University of Amsterdam, the Netherlands

- Hands-on experience in both cognitive modeling using evidence-accumulation models and cognitive neuroscience methods. Three-day introduction to evidence-accumulation modeling using Bayesian inference. Two days focused on how to use the evidence-accumulation framework to formalize performance in complex experimental paradigms, how to construct joint models of multiple tasks using

techniques such as hierarchical factor models, and how to combine evidence-accumulation and reinforcement-learning models.

- Lecturers: Prof. Andrew Heathcote, Prof. Dora Matzke, Dr. Michael Nunez, Dr. Steven Miletic, and Dr. Niek Stevenson

qHelp international seminar on Quantitative Psychology

Mar. - Apr. 2023

Arrifana, Portugal

- A one-week intensive seminar covering Shiny and R Markdown for interactive data reporting, Copula theory and its applications in dependence modeling, numerical methods, and optimization techniques for computational analysis.
- Lecturers: Julian Mollenhauer, Ana Lourenço, Prof. Luciano Stefanutti, Prof. Hans Colonius, Prof. Francis Tuerlinckx, and Prof. Sérgio Moreira

Statistical Rethinking

Jan. - Mar. 2023

Online Course

- Pre-recorded lectures and weekly meetings to discuss the material. This course teaches data analysis, focusing on scientific models: conceptual and causal models. Bayesian data analysis to connect scientific models to evidence.
- Lecturer: Prof. Richard McElreath

Fundamentals and Applications of Pupillometry

20 hours

University of Padova, Italy

- This course offers an introduction to the use of pupillometry in cognitive science, combining practical exercises in data processing with free software tools: OpenSesame for implementing classic cognitive psychology tasks and R for analyzing and visualizing continuous pupil diameter changes. Covering key topics such as noise sources, experimental design, data processing options, and the cognitive interpretation of pupillometric measures.
- Professor: Dr. Giulia Calignano

EEGLAB: analysis of electroencephalographic data

20 hours

University of Padova, Italy

- The EEGLAB and ERPLAB toolboxes, which operate within MATLAB, facilitate the analysis of electroencephalographic (EEG) data. This course offers foundational skills in data analysis and theoretical knowledge through practical examples, covering key topics such as data pre-processing and event-related potential (ERP) computation.
- Professor: Dr. Mattia Doro

Introduction to R

20 hours

University of Padova, Italy

- This course introduces the fundamentals of the open-source programming language R, teaching core programming concepts, data structures, and function creation through practical exercises to build versatile skills applicable across academic and professional settings.
- Professor: Dr. Filippo Gambarota

PsychoPy3: experimental programming

20 hours

University of Padova, Italy

- This course introduces participants to PsychoPy, an open-source Python-based software for designing and running behavioral science experiments, teaching them how to create, test, deploy, and manage experiments using its flexible and user-friendly graphical interface.
- Professor: Dr. Thomas Quettier

Scientific Societies Membership

AIP, Sezione Sperimentale	2022 - Present
Italian Reproducibility Network	2023 - Present
Cognitive Science Society	2024 - 2025
European Sleep Research Society	2023 - 2025
Society for the Improvement of Psychological Science	2022 - 2023
European Society for Cognitive Psychology	2021 - 2023

Languages

Italian	Mother Tongue
English	Proficient
Spanish	Beginner

Programming Languages

R / RStudio	Proficient
Stan	Intermediate
JAGS	Basic
Python	Basic
MATLAB	Basic

Research and Project Management Skills

PsychoPy/Pavlovia and OpenSesame/OSWeb	Intermediate
• Task development for behavioral research.	
GitHub and GitLab	Intermediate
• Version control and collaboration. • github.com/Mar-Cald, gitlab.pavlovia.org/mar.cald	
Docker	Basic
• Development and deployment of containers.	
Google Suite, Office Suite	Intermediate
• Gmail, Calendar, Drive, Meet, Docs, Forms, Sheets.	

R Packages

- *multibayes*: tools for Bayesian multiplicity adjustments (**maintainer**). mar-cald.github.io/multibayes.

Operating Systems

macOS	Intermediate
Windows	Intermediate
Linux	Intermediate

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